Initial EDA Report

Headtracking Measures as Indicators of Depressive Symptoms

360° VR Replication Study

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Dataset Overview & Demographics

Inclusion & Exclusion

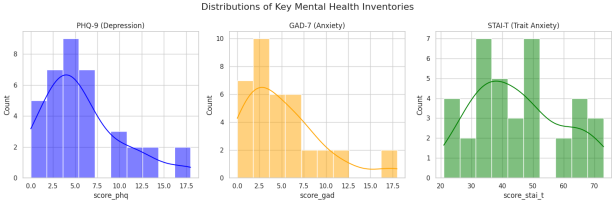
- Initial Sample: $N = 40$ college students.
- Exclusion: 3 participants removed due to significant simulator sickness (VRISE < 25).
- Final EDA Dataset: $N = 37$.

Demographics (N=37)

- **Age:** $M = 22.89$ ($SD = 1.82$), Range: 19–27.
- **Gender:** 33 Male / 4 Female.
- **VR Experience:** Generally low ($M = 1.41$ on a 2-point scale).

Mental Health Score Distributions

- **PHQ-9:** Right-skewed ($M = 5.81$). Sample is mostly in "Minimal" to "Mild" ranges.
- **GAD-7:** Matches depression skew ($M = 5.11$).
- **STAI-T:** Shows wider variance ($SD = 14.6$).

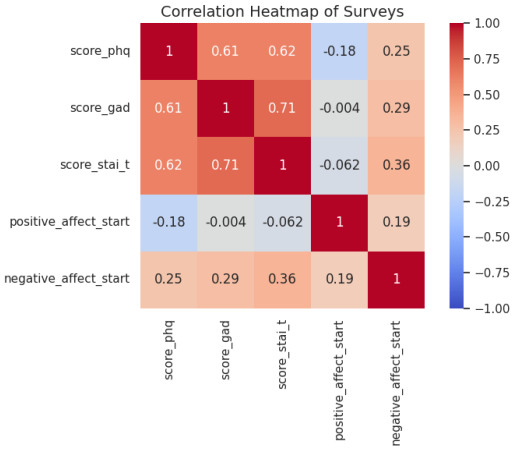


Variable Covariance Analysis

Key Correlations:

- **PHQ-9 & GAD-7:** Strong positive correlation ($r = 0.612$).
- **PHQ-9 & STAI-T:** Strong positive correlation ($r = 0.617$).

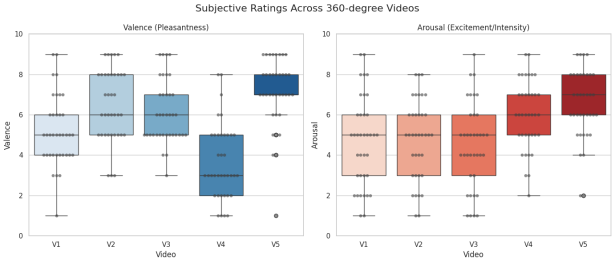
Takeaway: Anxiety must be treated as a covariate in future inferential models to isolate depressive psychomotor effects.



Subjective Affective Responses

Video Subjective Ratings

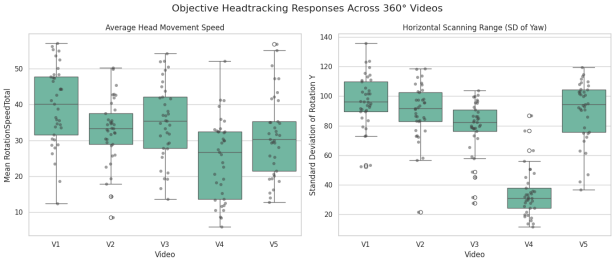
- **Valence:** V2 (Beach) and V5 (Tahiti) rated highly pleasant. V4 (Nun) rated highly unpleasant.
- **Arousal:** V4 elicited the highest arousal spikes.
- Emotional manipulation via 360° video environments was successful.



Objective Psychomotor Behavior by Video

Behavioral Shifts

- **V1 (Abandoned):** Highest active exploration (Speed $M = 39.6$).
- **V4 (Nun Horror):** Induced a "freezing" effect; lowest speed (24.8) and highly restricted Yaw scanning ($SD = 33.9$).
- **V2/V5 (Water):** High horizontal scanning, low vertical variance.



Controlling for False Discovery Rate (FDR) – Benjamini-Hochberg

- Instead of controlling the probability of making *even one* false positive (FWER), FDR controls the **expected proportion of false discoveries** among all significant results.
- Ideal for **exploratory science** (like this VR study), as it preserves statistical power while still keeping error rates in check.

The Benjamini-Hochberg (B-H) Procedure:

1. **Rank** all m obtained p-values from smallest to largest: $P_{(1)} \dots P_{(m)}$
2. **Calculate** the B-H critical value for each test: $\frac{i}{m} \times Q$
3. **Evaluate:** Find the largest p-value that is smaller than its corresponding critical value. That test, and **all tests ranked above it**, are deemed statistically significant.

Why not Bonferroni?

Bonferroni (α/m) is too stringent and assumes tests are independent. B-H provides a more nuanced, powerful threshold for highly correlated data (e.g., repeated headtracking measures).

Conclusion & Next Steps

Summary of EDA

1. Sample is strictly non-clinical but shows expected mental health score distributions.
2. Headtracking metrics (Speed, Pitch, Yaw) are highly sensitive to the VR environment presented.

Next Phases of Analysis

- **Repeated Measures ANOVA** to test for significant differences in headtracking across the different video environments.
- Apply the **Benjamini-Hochberg (B-H) correction** to all resulting p-values to safely control the False Discovery Rate across our multiple comparisons.